

allel data respectively. This result show that with similar number of parameters, our approach benefits from a better English pretrained model. When transferring from RoBERTa_{LARGE}, we obtain state-of-the-art results for five languages. It is worth mentioning that these models are still underfit and they have potential to push the XNLI performance further.

Currently, RAMEN only uses parallel data to initialize foreign embeddings. RAMEN can also exploit parallel data through translation objective proposed in XLM. We believe that by utilizing parallel data during the fine-tuning of RAMEN would bring additional benefits for zero-shot tasks. We leave this exploration to future work. In summary, starting from BERT_{BASE}, our approach obtains competitive bilingual LMs with mBERT for zero-shot XNLI. Our approach shows the accuracy gains when adapting from a better pretrained model.

4.2 Universal Dependency Parsing

We build on top of RAMEN a graph-based dependency parser (Dozat and Manning, 2016). For the purpose of evaluating the contextual representations learned by our model, we *do not* use part-of-speech tags. Contextualized representations are directly fed into Deep-Biaffine layers to predict arc and label scores. Table 2 presents the Labeled Attachment Scores (LAS) for zero-shot dependency parsing.

	🇬🇧	fr	vi	zh	ru	ar	hi	avg
mBERT	🇬🇧	71.6	35.7	26.6	65.2	36.4	30.4	44.3
RAMEN _{BASE}								
+ BERT	🇬🇧	78.0	38.3	30.1	67.2	40.6	38.6	48.8
	🇬🇧	77.0	36.7	30.8	66.8	40.9	40.9	48.9
+ RoBERTa	🇬🇧	79.1	38.9	31.5	67.7	42.2	41.2	50.1
	🇬🇧	78.5	39.4	31.3	65.1	41.4	44.2	50.0
RAMEN _{LARGE}								
+ BERT	🇬🇧	78.2	39.5	30.5	65.6	45.3	44.0	50.5
	🇬🇧	78.9	38.4	30.5	66.3	43.4	44.1	50.3
+ RoBERTa	🇬🇧	79.7	40.0	32.2	65.7	44.1	44.6	51.1
	🇬🇧	79.1	39.2	30.5	65.4	43.8	46.8	50.8

Table 2: LAS scores for zero-shot dependency parsing. 🇬🇧 indicates parallel data is used for initialization. Punctuation are removed during the evaluation. The best results are marked in **bold**.

We first look at the fairest comparison between mBERT and monolingually initialized RAMEN_{BASE}+BERT. The latter outperforms the former on all the languages with the average of 4.5 LAS point. We observe the largest gain of +8.2 LAS point. We observe the largest gain of +8.2 LAS point. We observe the largest gain of +8.2 LAS point for Hindi and +6.2 LAS for French. Arabic enjoys

+4.2 LAS from our approach. With similar architecture (12 or 24 layers) and initialization (using monolingual or parallel data), RAMEN+RoBERTa performs better than RAMEN+BERT for most of the languages. Arabic and Hindi benefit the most from bigger models. For the other four languages, RAMEN_{LARGE} renders a modest improvement over RAMEN_{BASE}.

5 Analysis

Throughout this section, we carry out the main analysis for the RAMEN_{BASE}+BERT model that only use monolingual data.

5.1 How does linguistic knowledge transfer happen through each training stages?

We evaluate the performance of RAMEN+BERT_{BASE} (initialized from monolingual data) at each 20K training steps. The results are presented in Figure 2.

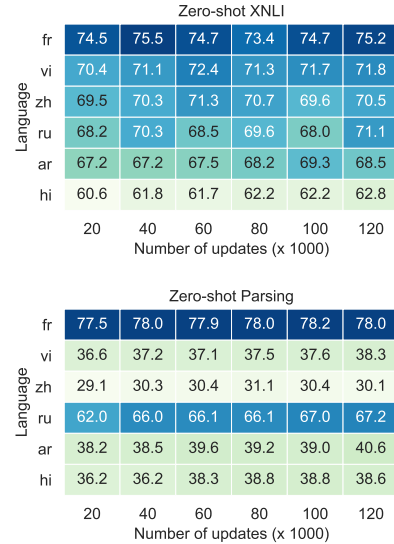


Figure 2: Accuracy and LAS evaluated at each checkpoints.

We observe the general trend that further fine-tuning RAMEN improves the performances of two zero-shot tasks. Just after 40,000 fine-tuning updates—corresponding to seven training hours—RAMEN_{BASE}+BERT surpasses mBERT on both XNLI (Table 1) and universal dependency parsing (Table 2). Interestingly, we find that the English encoder can quickly adapt to the foreign syntax that has different word order (ar and hi) with a little training data.

Language similarities seem to have more impact on transferring syntax than semantics. French en-